# Heitor Baldo, PhD

*Applied Mathematician / Researcher*

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*heitorbaldo*

## Academic Appointments

2024 – 2026 **Postdoctoral Research Fellow**, *University of São Paulo*, Brazil.

2024 – 2025 **Visiting Postdoctoral Research Fellow**, *Leipzig University*, Germany.

2019 – **Affiliate Researcher**, *IGDORE Institute*, Sweden.

## Education

2019 – 2024 **Ph.D. in Bioinformatics (Mathematical Neuroscience)** , *University of São Paulo*, Brazil.
Thesis: *Towards a Quantitative Theory of Digraph-Based Complexes and its Applications in Brain Network Analysis*, advised by Koichi Sameshima and co-advised by André Fujita.

2014 – 2016 **M.S. in Applied and Computational Mathematics**, *University of Campinas*, Brazil.
Thesis: *Álgebras de Clifford e de Cayley-Dickson*, advised by Jayme Vaz Jr.

2009 – 2013 **B.S. in Mathematics**, *University of Campinas*, Brazil.

## Publications

### Preprint Manuscripts

2026 **Baldo, H., Baccalá, L., Fujita, A., & Sameshima, K.**, *Directed Q-Analysis and Directed Higher-Order Connectivity on Digraphs: A Quantitative Approach*, arXiv:2605.14178, `https://doi.org/10.48550/arXiv.2605.14178`
(preprint - submitted to a peer-reviewed journal)

### Under Review Manuscripts

2026 **Baldo, H. & Fujita, A.**, *A Temporal Hypergraph Motif Framework for the Analysis of Higher-Order Dynamics of Epileptic Brain Networks*, (under review).

### Unpublished Manuscripts and Academic Writings

2023 **Baldo, H.**, *Notes on Simplicial Neural Networks for Path Complexes*, (research notes).

2022 **Baldo, H.**, *Notes on Collapses and Persistent Path Homology*, (unpublished manuscript).

## Open-Source Software

Since 2022 **Research Packages: DigplexQ, TemporalHypermotifs, GeometrySpectralEvo, PyTropical, Pytroids**, `https://github.com/heitorbaldo?tab=repositories`
- Open-source Python packages providing computational tools for research in directed higher-order network analysis and directed Q-analysis, temporal graphs and hypergraphs, tropical mathematics, and graphic matroid theory. Tools: Python, Sphinx, PyPi, NumPy, SciPy, Pandas, NetworkX, HyperNetX, XGI, Gudhi, Giotto-TDA, Numba.

## Talks

2025 "Spectral Density of Multilayer Hypergraphs and Applications" in *6th Central German Meeting on Bioinformatics*, Leipzig University.

2022 "Graph Cellular Automata and Beyond: Applications in Network Neuroscience." Institute of Mathematics and Statistics - USP. (slides)

2022 "Path Homology Theory of Digraphs and Its Applications to Brain Network Analysis" in *Universitä Leipzig - USP Workshop 2022* at the Institute of Mathematics and Statistics - USP. (slides)

## Teaching Experiences

Spring 2021 **Teaching Assistant**, *University of São Paulo*, Brazil.
Course: *Multivariate Data Analysis* (MAE0330).

Winter 2017 **Lecturer**, *University of Campinas*, Brazil.
Course: *Complex-Valued Neural Networks* (Minicourse).

Winter 2016 **Lecturer**, *University of Campinas*, Brazil.
Course: *Bio-Inspired Algorithms: An Introduction* (Minicourse).

Fall 2016 **Teaching Assistant**, *University of Campinas*, Brazil.
Course: *Linear Algebra* (MA327).

## Technical Skills

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|---|---|
| Programming | Python • Julia • R • C/C++ • Rust • SQL • TypeScript • HTML/CSS |
| ML/DL | PyTorch • TensorFlow • scikit-learn • XGBoost • PyTorch Geometric • DGL • JAX |
| Backend & APIs | FastAPI • Pydantic • Uvicorn • PostgreSQL • Redis • SQLAlchemy • REST |
| Frontend | Streamlit • Gradio • React • Next.js • TypeScript |
| MLOps & Infra | Docker • MLflow • Weights & Biases • DVC • Evidently AI • Prometheus • Grafana |
| HPC & Parallel | CUDA • CuPy • Numba • Dask • Ray |
| Math & Neuro | SageMath • MATLAB/Octave • EEGLAB • Brainstorm • FreeSurfer • SPM • FieldTrip • fMRIPrep • Brian2 • MNE |
| Others | Git • Linux • pytest • TDD • CI/CD • Sphinx • Agile • LaTeX |

## Selected Participation in Events

2026 Homotopy Theory, K-theory, and Topological Data Analysis. *Fields Institute.*

2025 Thirteenth Symposium on Compositional Structures (SYCO 13). *University of Cambridge.*

2025 Workshop "Biology and Geometry". *Max Planck Institute for Mathematics in the Sciences.*

2025 6th Central German Meeting on Bioinformatics. *Leipzig University.*

2025 40th TBI Winterseminar in Bled. *Bled, Slovenia.*

2024 Twelfth Symposium on Compositional Structures (SYCO 12). *University of Cambridge.*

2021 Workshop on Algebraic Graph Theory and Quantum Information. *Fields Institute.*

2021 4th Workshop on Algebraic Graph Theory and its Applications. *Mathematical Center in Akademgorodok.*

2020 XLIII Annual Meeting of the Brazilian Society for Neuroscience and Behavior. *Online.*

2020 Seminars on Probability and Stochastic Processes. *University of São Paulo.*

2017 6th Brazilian Conference on Intelligent Systems. *Federal University of Uberlandia.*

2016 II Brazilian Congress of Young Researchers in Pure and Applied Mathematics. *University of Campinas.*

2015 IV School and Workshop on Lie Theory. *University of Campinas.*

2014 Workshop Many Faces of Distances. *University of Campinas.*

## Certifications & Training

Spring 2026 Introduction to Agent-Based Modeling. *Santa Fe Institute.* [Certification]

Spring 2018 Statistical Learning. *Stanford Online – Stanford University.* [Certification]

Spring 2017 Introduction to Complexity. *Santa Fe Institute.* [Certification]

Spring 2017 Minicourse on Machine Learning for Many-Body Physics. *IFT-Unesp.*

## Languages

- **Portuguese:** Mother Tongue.
- **English:** Advanced Level (C1*).
- **German:** Pre-intermediate Level (A2*)

(*) *Common European Framework of Reference (CEFR) level.*

Last updated August 18, 2026.